ERIN E. FLOWERS

ASTROPHYSICIST & ASSISTANT DIRECTOR OF STEM EDUCATION

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EDUCATION

PhD Astrophysical Sciences, Princeton University
September 2017 – May 2023
- Thesis: *Titan's Anomalous Atmosphere from Top to Turf*

MA Astrophysical Sciences, Princeton University
September 2017 – May 2019

BA Astrophysics, Columbia University
September 2013 – May 2017
- Thesis: *Using High Resolution Spectroscopy to Determine the Rotation Rate of Hot Jupiter Exoplanets*

Major GPA: 3.80/4.0 Overall GPA: 3.72/4.0

WORK & RESEARCH EXPERIENCE

Assistant Director
Princeton University, Council on Science and Technology *July 2023 – present*
- Conduct cutting-edge scientific research in personal field of study (astrophysics)
- Oversee STEM course content development across all university STEM departments
- Develop and teach university STEM courses
- Mentor graduate and undergraduate students in STEM research
- Develop and execute programming to promote STEM literacy on and off campus

Research Associate
NASA Dragonfly Mission *January 2022 – present*
- Spearheaded efforts to improve simulations of Titan's atmospheric dynamics with the goal of informing larger mission collaboration's engineering efforts
- Develop and optimize code in Python, IDL, and Fortran to solve the three-dimensional equations that govern atmospheric and surface dynamics, orbital dynamics, and atmospheric chemistry utilizing numerical integrators, finite difference methods, and parallelized computing on the NASA Pleiades Supercomputer
- Analyzed and compared simulation results with extant observational data to improve understanding of atmosphere-surface interactions on Titan

Graduate Research Assistant
Princeton University, Astrophysical Sciences Department *September 2017 – May 2023*
- Spearheaded the exploration of Titan's atmospheric dynamics, chemistry, and interactions with its space environment over long timescales
- Conducted statistical analysis of known exoplanet characteristics and managed follow-up field observations
- Conceptualized, coded, and executed simulations of multidimensional dynamic and chemical systems in Python, IDL, Fortran, and C, utilizing numerical integrators, finite difference methods, and parallelized computing optimization
- Published results in peer-reviewed journals and presented results at national and international conferences

TECHNICAL SKILLS

Programming Languages	Proficient in Python, IDL, Fortran, C, C++, HTML, Linux, bash; Familiar with SQL
Libraries & Packages	NumPy, SciPy, AstroPy, pandas, xarray, matplotlib, seaborn, bokeh, scikit-learn
Modeling & Computational	Numerical integrators, iterative methods, Monte Carlo methods, AI-assisted coding, parallelized computing, statistical & machine learning

Mathematical Techniques	Calculus of variations, PDE, ODE, linear algebra, statistics, probability, optimization

SELECT PUBLICATIONS

1. **Flowers, E.**, Newman, C., Lian, Y., Cordiner, M., Nixon, C., Thelen, A., Cosentino, R., Richardson, M., Lee, C., and Toigo, A., "Reproducing Superrotation in Titan's Stratosphere with the TitanWRF GCM" in review 2024
2. **Flowers, E**, "Titan's Anomalous Atmosphere from Top to Turf" (PhD thesis, Princeton University, 2023)
3. **Flowers, E.** and Chyba, C., "Shock Synthesis of Organic Molecules by Meteors in the Atmosphere of Titan" (2023) doi: /arXiv: 2307.10293
4. **Flowers, E.**, Brogi, M., Rauscher, E., Kempton, E. M.-R., & Chiavassa, A., "The High-Resolution Transmission Spectrum of HD 189733b Interpreted with Atmospheric Doppler Shifts from Three-Dimensional General Circulation Models" doi: /arXiv:1810.06099

h-index/citation count: 9/349

SELECT PRESENTATIONS

1. **Flowers, E.**, Newman, C., Lian, Y., Cordiner, M., Nixon, C., Thelen, A., Cosentino, R., Richardson, M., Lee, C., and Toigo, A., "Reproducing Stratospheric Superrotation in Titan's Atmosphere with the TitanWRF GCM" – Extreme Solar Systems V, New Zealand, 2024
2. **Flowers, E.** and Chyba, C., "Shock Synthesis of Organic Molecules by Meteors in the Atmosphere of Titan" – American Astronomical Society Winter Meeting 2023
3. **Flowers, E.**, "Titan: Why it's the Coolest Place in the Solar System & Past and Future Missions to Prove it" – Indiana University Physics Colloquium 2023
4. **Flowers, E.** and Chyba, C., "Energy Delivery Via Meteors Into Titan's Atmosphere" – American Geophysical Union Fall Meeting 2022
5. **Flowers, E.**, Brogi, M., Rauscher, E., M-R Kempton, E., and Chiavassa, A., "Using Transmission Spectroscopy to Determine the Rotation Rate of HD 189733b" – University of Chicago Astronomy & Astrophysics Colloquium 2020

AWARDS & FELLOWSHIPS

- Princeton University Association of Black Graduate Alumni Patrice Y. Johnson *80 Service Award (2023)
- Princeton University Prison Teaching Initiative Math & Physics Fellowship (2018 - 2023)
- Princeton University Access, Diversity and Inclusion Graduate Fellowship (2018 - 2023)
- Princeton University Graduate Student Teaching Award (2021)
- National Science Foundation Graduate Research Fellowship (2017 - 2022)
- Columbia University Named Scholarship (2015 - 2016)
- American Physical Society Minorities in Physics Scholarship (2013 - 2015)